Operational Amplifier Designs: 5-OTA Version and Folded Cascode Version

Xianduo Yang
School of Microelectronic
Southern University of Science and Technology
ShenZhen, China
12311806@mail.sustech.edu.cn

Xiaojun Zhou
School of Microelectronic
Southern University of Science and Technology
ShenZhen, China
12311815@mail.sustech.edu.cn

Quan Pan
School of Microelectronic
Southern University of Science and Technology
ShenZhen, China
panq@mail.sustech.edu.cn

***Abstract*—This project designs two differential-input single-output operational amplifiers in a 0.18μm CMOS process to meet specified performance requirements and ±10% VDD (1.8 V) fluctuation stability. The two topologies are a two-stage 5-OTA with RC frequency compensation and a single-stage folded cascode. Simulation results show both meet all requirements: the two-stage 5-OTA achieves 67.99–68.93 dB differential gain, 162.5–164.4 MHz unity gain frequency, 24.8–35.4 V/μs slew rate, and 264.5–325.7μW power consumption. The single-stage folded cascode delivers 46.89–68.52 dB differential gain, 96.78–101.7 MHz unity gain frequency, 24.7–53.4 V/μs slew rate, and 235.8–288.8μW power consumption. Both verify feasibility for high-performance analog applications.**

Keywords—analogue CMOS circuit design, amplifier, gain, phase margin, slew rate, OTA, folded cascode structure

I. Introduction

Operational amplifiers(Op-amp) are important in analogue circuits, which has high input impedance and open loop gain, while presenting a low output impedance. Ideally, an Op-amp has infinity input impedance and open loop gain, zero output impedance and infinity bandwidth. It is commonly used in analogue circuits with negative feedback.

II. Project Requirement and Results

In this project, it is required to design a differential input and single output operational amplifier with the following requirements shown in table 1. In addition, if V_{DD} fluctuates ±10%, the circuit can still match the demands, or use a single stage amplifier to achieve the requirements, a bonus will be granted. It should be noticed that the power consumption in the circuit should be lower than 500uW, excluding the bias current power.

TABLE I. Project Requirements

Process	0.18 μm
V_{DD}	1.8 V
V_{SS}	0 V
Load Cap	1 pF
A_{DM0} (low-frequency open-loop small-signal gain)	≥ 1000 V/V
A_{CM0} (low-frequency open-loop small-signal common-mode gain)	≤ 0.1 V/V
Phase margin	≥ 60°
Unity gain frequency	≥ 100 MHz
Slew rate	≥ 10 V/μs
Output voltage swing (differential peak to peak)	≥ 800 mV
Power consumption	Minimum

In this project we have achieved all the basic requirements and satisfied the bonus requirements. The following tables show our circuit properties, including a two-stage 5-OTA structure amplifier and a folded cascoded structure.

TABLE II. Two Stages 5-OTA Amplifier Properties

Requirements	VDD=1.62V	VDD=1.8V	VDD=1.98V
A_{DM0}	67.99dB	68.52dB	68.93dB
A_{CM0}	-20.52dB	-20.27dB	-20.03dB
f_T	162.5MHz	163.5MHz	164.4MHz
PM	67.03°	66.76°	66.60°
Slew Rate (Up/Down)	24.8V/us 35.3V/us	24.8V/us 35.4V/us	24.8V/us 35.4V/us
Swing	800.12mV	800.13mV	800.11mV
Total Power	264.5uW	295.1uW	325.7uW

TABLE III. Single Stage Folded Cascode Amplifier Properties

Requirements	VDD=1.62V	VDD=1.8V	VDD=1.98V
A_{DM0}	46.89dB	68.52dB	64.86dB
A_{CM0}	-24.95dB	-24.90dB	-24.96dB
f_T	96.78MHz	101.1MHz	101.7MHz
PM	74.20°	75.46°	75.67°
Slew Rate (Up/Down)	24.7V/us 53.4V/us	26.7V/us 49.3V/us	28.2V/us 48.5V/us
Swing	800.6mV	801.3mV	801.4mV
Total Power	235.78uW	262.40uW	288.80uW

As shown above, all the requirements are spontaneously met, including the bonus part.

III. Basic Parameters

Before starting our circuit design, there is a critical issue that we need to know some key parameters, especially Vth, μ_n, C_{ox} for both PMOS and NMOS. However, it is hard to find the exact number in the gpdk process, and hence, we created a simple MOS circuit schematic and launch DC simulation to calculate these parameters. The circuit is shown in figure 1.

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

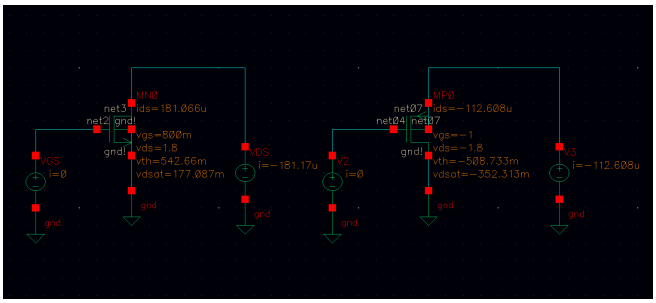


FIGURE 1. BASIC MOS CIRCUIT

In this circuit, we set both PMOS and NMOS W=2u and L=0.18u, and we apply $V_G = 0.8V, V_{DD} = 1.8V$. After DC simulation, by observation, $V_{th,n} \approx 0.55V, V_{th,p} \approx -0.51V$, and then we use the equation:

$$I_{d,n} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{th,n})^2$$

$$I_{d,p} = \frac{1}{2}\mu_p C_{ox}\frac{W}{L}(V_{SG} - |V_{th,p}|)^2$$

We can get $\mu_n C_{ox} = 3.0 \times 10^{-4} F \cdot V^{-1} \cdot s^{-1}$ for NMOS and $\mu_p C_{ox} = 7.8 \times 10^{-5} F \cdot V^{-1} \cdot s^{-1}$ for PMOS. Although we neglect the second order effects, and they are just an approximation, we can still use them to design our circuit draft and we can modify our circuit based on the draft. These parameters are global parameters, crucial for amplifier design.

IV. TWO STAGES 5-OTA AMPILIFER DESIGN

The design approach can be summarized below, which contains the current mirror design, first stage amplifier design, second stage amplifier design and frequency compensation design.

A. NMOS Current Mirror design

Since the required more than 60dB open loop gain is hard to implement using only one stage amplifier, during the early stage of our design, we would like to make a two-stage amplifier. First, we need to determine the power usage, after discussing, we attempt to lower our power consumption in the first stage, for instance, 54uW for the first stage and 180uW for the second stage. In terms of the bias current, the first stage needs 30uW, and the second stage needs 100uA. If we set the current source to 20uA, the current mirror W/L ratio should be 2:3:10. In order to make our calculations simpler, we set all the transistors L=1um. The W in the current mirror is 2.67um, 4um, 13.3um, respectively. After that we need to decide each W in the rest of the transistors. Here's our hand calculations draft, shown in figure 2.

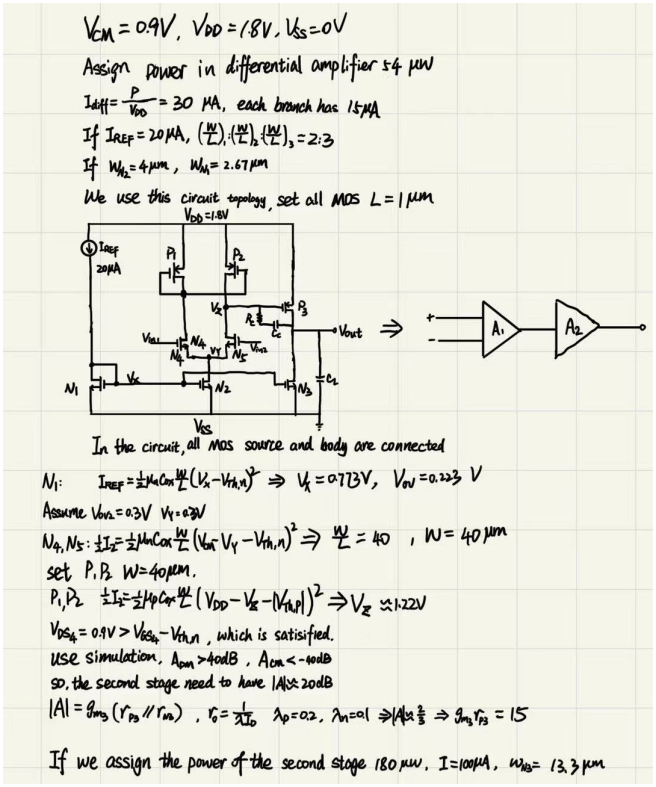


FIGURE 2. HAND CALCULATION DRAFT

B. First Stage Ampilifer Design

In our calculations, the basic formulas are still the same, which are:

$$I_{d,n} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{th,n})^2$$

$$I_{d,p} = \frac{1}{2}\mu_p C_{ox}\frac{W}{L}(V_{SG} - |V_{th,p}|)^2$$

We first calculated the current mirror's gate voltage, which is 0.773V. later, we calculated the NMOS differential pairs, which is 40um width, we assume the top PMOS current mirror width to be 40um, and we get the input DC voltage for the second stage amplifier is about 1.22V. Put the first stage amplifier into cadence, after DC and AC simulation, fortunately, all the transistors are saturated, and A_{DM0}> 40dB while A_{CM0}< -40dB, shown in figure 3 and 4, and hence the second Stage amplifier needs about 20dB gain and the total over 60dB differential mode gain and below -20dB common mode gain can be seen theoretically.

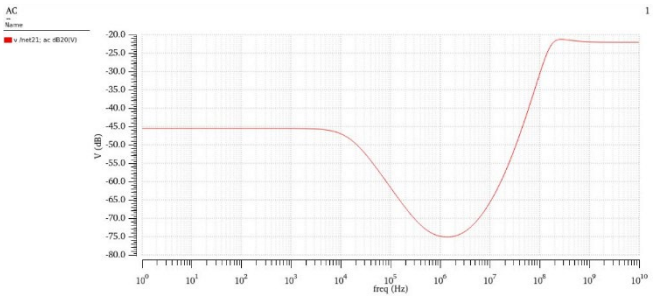


FIGURE 3. FIRST STAGE COMMON MODE GAIN

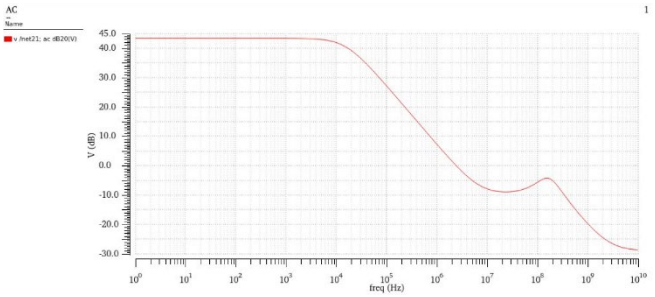


FIGURE 4. FIRST STAGE DIFFERENTIAL MODE GAIN

C. *Second Stage Ampilifer Design*

To find the second stage PMOS parameter, we scanned the parameter. In a new schematic shown below, we set the outside environment of the second stage amplifier the same as the overall schematic.

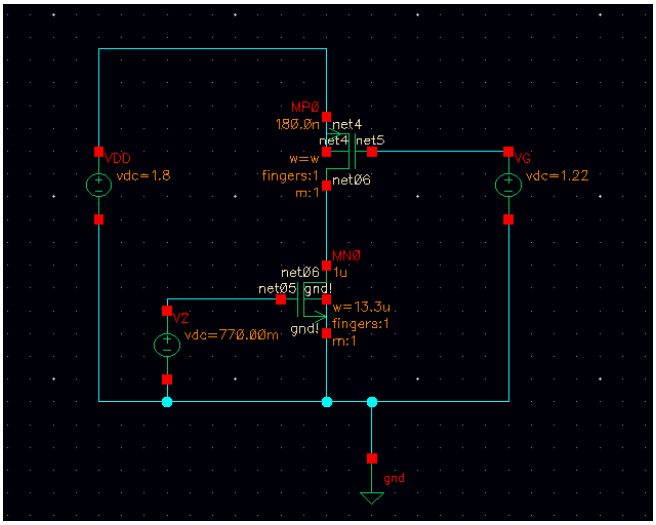


FIGURE 5. THE CIRCUIT SCHEMETIC TO FIND W

We set L=0.18um to accommodate more choice of W. Since the second stage gain should be -g_m ($r_{0,p}||r_{0,n}$), and the equation for r_0 is

$$r_0 = \frac{1}{\lambda I_D}$$

The typical λ value for PMOS is 0.2, and the typical λ value for NMOS is 0.1, and hence, the gain can be rewritten as $\frac{2}{3} g_m \cdot r_{0,p}$. The method we use is to observe the intrinsic gain of the PMOS, and a 20dB gain corresponds to an intrinsic gain equals to 15 . The figure showed the result of scanning.

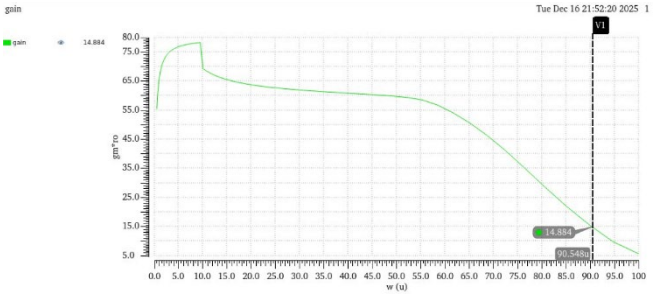


FIGURE 6. THE SCANNING RESULT

The figure shows that in order to achieve our goal, the PMOS width should be 90um. Now all the MOS sizes are uniquely determined. However normally the bandwidth is not satisfied and hence a frequency compensation is a demand.

D. *Frenquency Compensation*

If there is no frequency compensation, there will exist poles and zeros near GBW region, and the circuit may encounter stability and bandwidth issue. The best choice is to make a frequency compensation. In this circuit, we use a resistor and a capacitor connect in series between the output of the first and the second stage. We set the capacitance C_C to 1pF. To move the zero well into the left half plane to cancel the first non-dominate pole, the resistance is obtained by the following equation:

$$R \approx \frac{C_L + C_C}{g_{m,p} C_C}$$

We could get gm by cadence DC simulation, showed in the figure below. From the figure, $g_{m,p}$ =2.175mS.

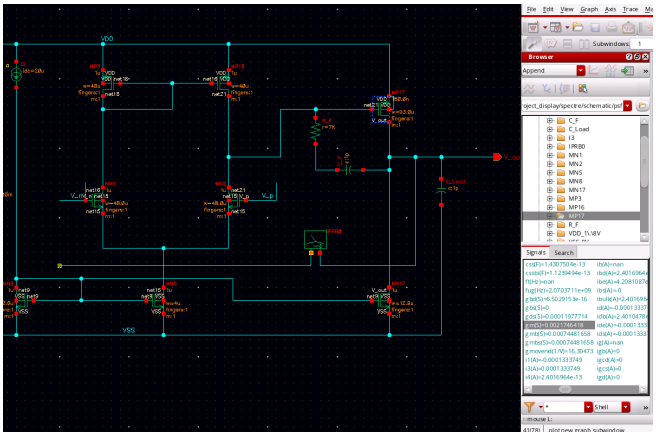


FIGURE 7. GET GM OF THE PMOS

CL is Defined by the requirement. According to the equation above, to perfectly compensate, the Resistance of the resistor should be 920Ω.

Until now, all the parameters are get. The draft is completed.

E. *Modification and Bonus*

It is important to know that all the above calculations neglect almost ALL second order and even higher effects, including the channel length modulation, mobility degradation, DIBL, and more, however, the reality will not neglect them. These effects potentially pose threats to the performance and stability in our circuit. The circuit using the above parameters will doomed to be imperfect, including the unity gain phase, phase margin, output voltage swing and more, and hence some small modifications are needed. Based on the draft circuit, after trials and errors, we finally get the ultimate result, concluded in the table below. Using these parameters, all the requirements are contended.

TABLE IV. TWO STAGES 5-OTA AMPLIFIER PARAMETERS

Item	Parameter
I_{REF}	20uA
N1	W=2.6um, L=1um
N2	W=4um, L=1um

N3	W=12.9um, L=1um
N4, N5	W=48um, L=1um
P1, P2	W=40um, L=1um
P3	W=93um, L=0.18um
CC	1pF
RC	7kΩ
CL	1pF

In the requirement, if VDD fluctuates ±10%, the circuit can still satisfy the requirement, a bonus can be granted. Fortunately, by verification, without conducting any changes in the circuit, it can meet this requirement.

V. Two Stages 5-OTA Ampilifer Results

The overall schematic of the circuit is shown below, which clearly display all the design parameters.

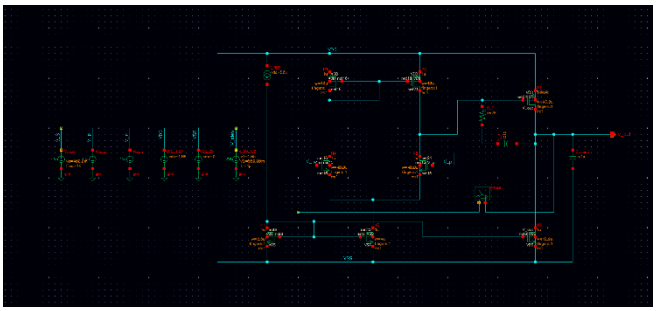


FIGURE 8. THE OVERALL CIRCUIT SCHEMATIC

Use the simulation tool ADEL, we can simulate our circuit, and the following figures show the result and performance of the two-stage 5-OTA based operational amplifier, under 1.8V 1.62V(1.8 × 90%) and 1.98V(1.8 × 110%) V_{DD}.

A. Under 1.8V V_{DD}

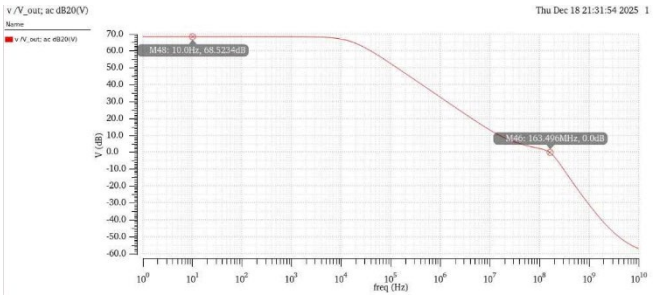


FIGURE 9. THE DIFFERENTIAL MODE GAIN AND UNITY GAIN FREQUENCY

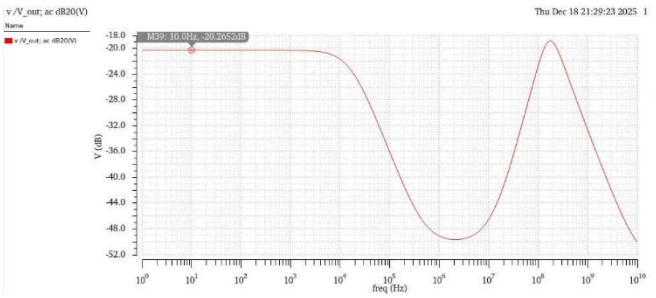


FIGURE 10. THE COMMON MODE GAIN

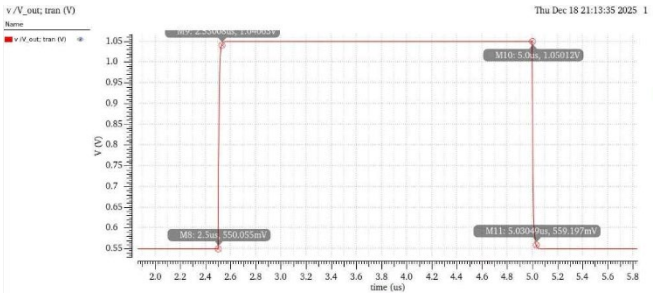


FIGURE 11. THE SLEW RATE

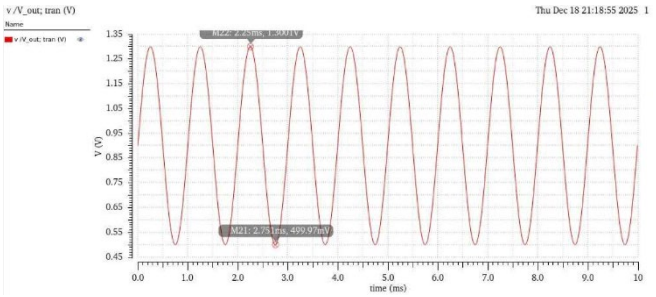


FIGURE 12. THE OUTPUT VOLTAGE SWING

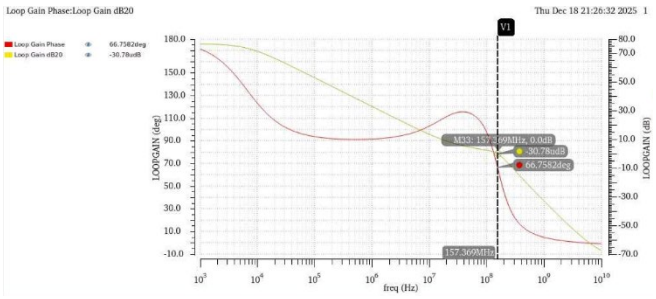


FIGURE 13. THE PHASE MARGIN

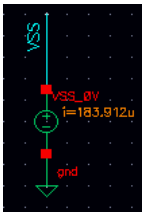


FIGURE 14. THE POWER CONSUMPTION WITH 20UA IREF

B. Under 1.62V V_{DD}

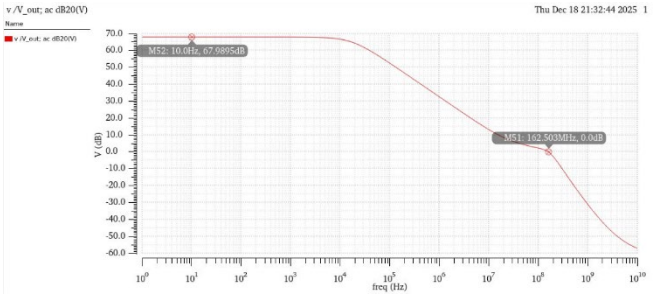


FIGURE 14. THE DIFFERENTIAL MODE GAIN AND UNITY GAIN FREQUENCY

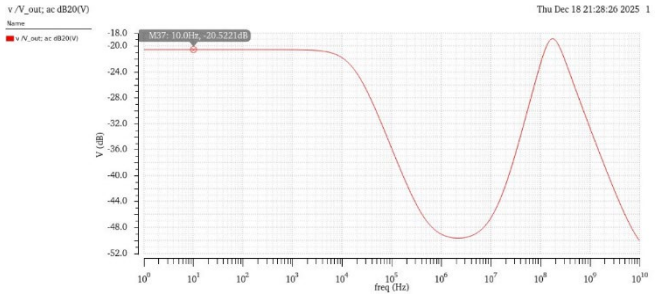


FIGURE 15. THE COMMON MODE GAIN

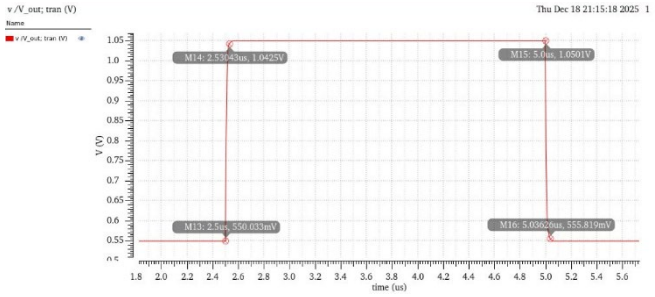


FIGURE 16. THE SLEW RATE

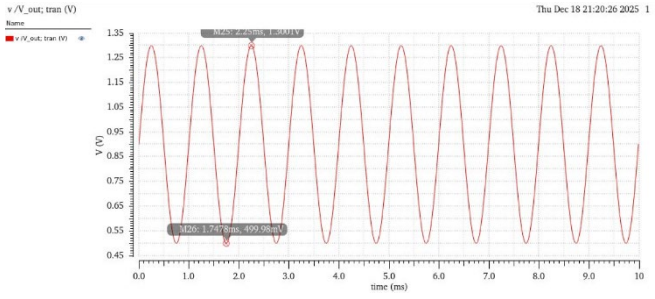


FIGURE 17. THE OUTPUT VOLTAGE SWING

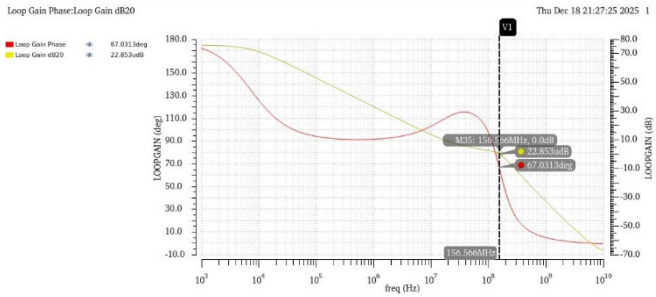


FIGURE 18. THE PHASE MARGIN

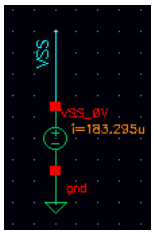


FIGURE 19. THE POWER CONSUMPTION WITH 20UA IREF

C. *Under 1.98V V_{DD}*

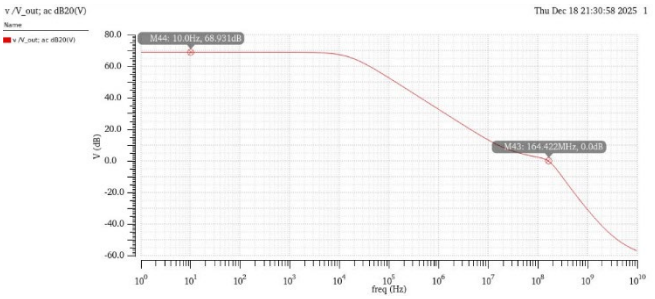


FIGURE 20. THE DIFFERENTIAL MODE GAIN AND UNITY GAIN FREQUENCY

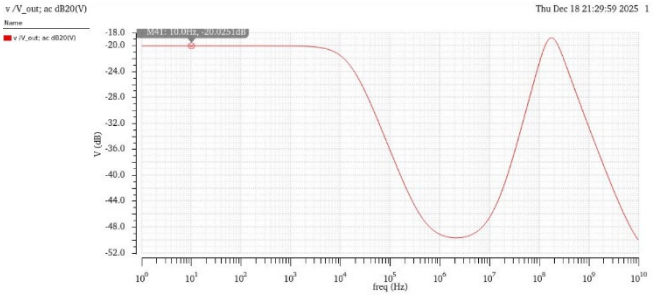


FIGURE 21. THE COMMON MODE GAIN

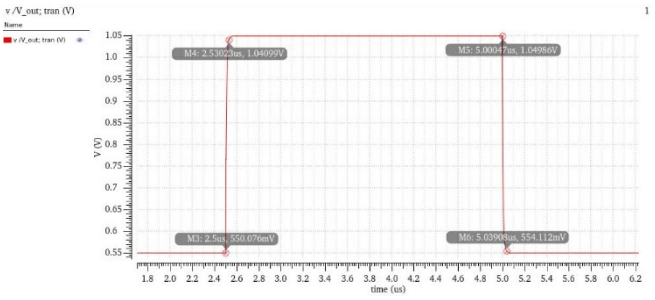


FIGURE 22. THE SLEW RATE

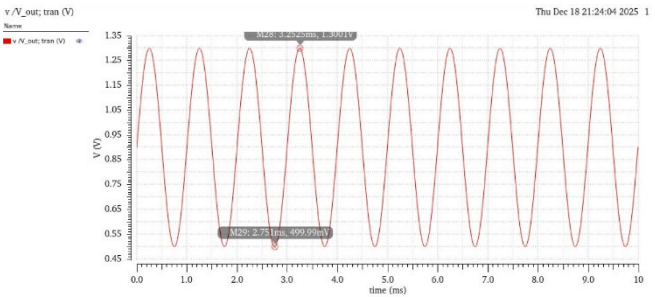


FIGURE 23. THE OUTPUT VOLTAGE SWING

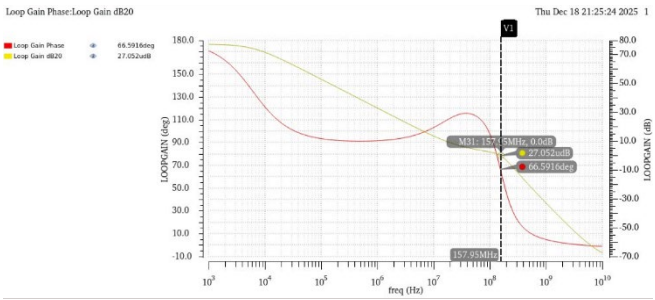


FIGURE 24. THE PHASE MARGIN

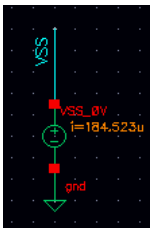


FIGURE 25. THE POWER CONSUMPTION WITH 20UA IREF

As shown above, this circuit performs well under the given conditions. If the reader wants to see them more clearly, table II is a brief conclusion of all the performance above.

VI. SINGLE STAGE FOLDED CASCODE AMPLIFIER DESIGN

The five-transistor OTA configuration offers a simple structure but suffers from relatively low gain. To meet the design specifications, a two-stage amplifier circuit is typically required. Therefore, if we aim to achieve the design requirements with a single-stage topology, we must consider using a telescopic cascode amplifier, which provides higher gain.

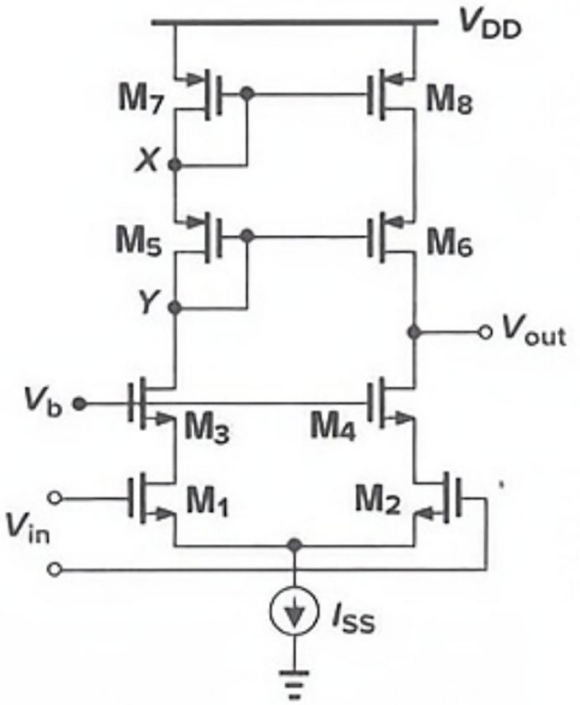


FIGURE26 . THE TELESCOPE STRUCTURE

However, the single-stage telescopic cascode amplifier has a drawback: the series stacking of five transistors from V_DD to ground results in a significant voltage drop, severely limiting the output swing. In most cases, it also needs to be paired with a two-stage configuration to achieve a larger output swing.

In contrast, the folded cascode amplifier reduces one transistor layer compared to the telescopic topology, thereby offering a more generous output swing. The tradeoff is that the folded structure requires additional current to operate. Consequently, we subsequently attempt to design a single-stage folded cascode amplifier to meet the design requirements.

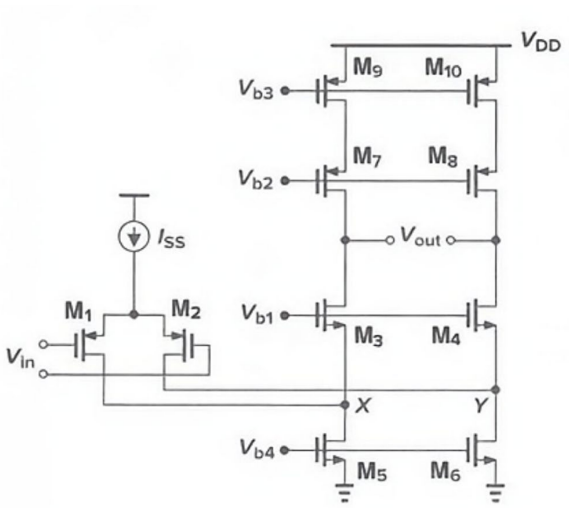


FIGURE 27. THE FOLDED CASCODE STRUCTURE

A. The First Attempt

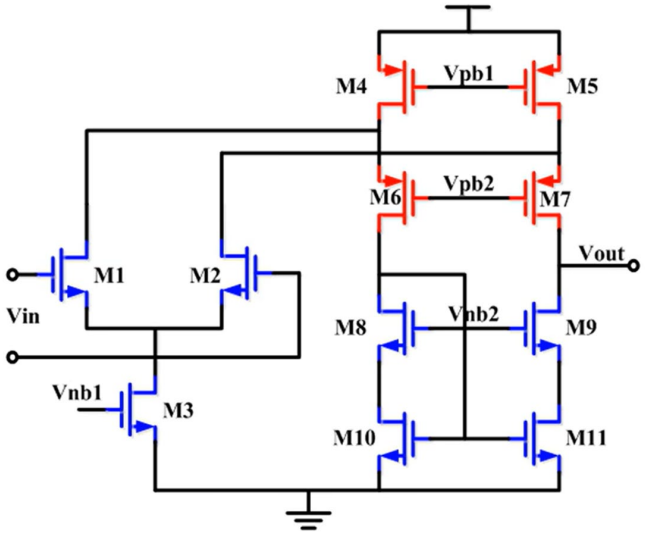


FIGURE 28. THE FOLDED CASCODE STRUCTURE

As specified in Table 1, the design requirements are: Slew rate≥ $10V/\mu s$ and load capacitance $C_L = 1pF$. Based on the definition of slew rate and the relationship between capacitor charging/discharging, the following formula is derived:

$$I = SR \cdot C_L$$

Substituting the values, we get $I_3 \geq 10\mu A$. To avoid cascode zero current, we typically design the current matching as $I_4 = I_5 = 1.2I_1 = 1.2I_2 = 0.6I_3$, leading to the total current $I_{total} = 2.2I_3$.

Next, an initial target power consumption was set to guide subsequent design and determine transistor parameters. At a supply voltage of 1.8 V, the target power consumption was set to $200\mu W$. Excluding the current sources, the total current is approximately $110\mu A$, thus yielding $I_3 = 50\mu A$.

According to the relationship between transconductance, load capacitance, and the designed bandwidth:

$$g_{m1} \approx 2\pi \cdot GB \cdot C_L$$

The transconductance of the differential pair is calculated as $g_{m1} = g_{m2} \approx 630\mu S$, Using the transistor aspect ratio formula:

$$\frac{W}{L} = \frac{g_m}{\mu_n C_{ox} V_{od}}$$

When setting the overdrive voltage $V_{OV} = 200mV$, we obtain $\frac{W}{L}_{1,2} \approx 10$. For the 180 nm process, to mitigate short-channel effects, L is generally chosen as $(2\sim6) \times 180\ nm$ Here, $L = 1\mu m, W = 10\mu m$ were selected.

For the folded cascode amplifier, the output swing range is given by:

$$2V_{OVn} < V_{out-folded} < V_{DD} - 2V_{OVp}$$

With $V_{DD} = 1.8V$ and considering the swing design requirements, we set $V_{OV8-9} = 250mV$, $V_{OV4-7} = 200mV$. The estimated swing margin is approximately 0.9 V, which meets the design requirements. Using the aspect ratio formula for current-biased transistors:

$$\frac{W}{L} = \frac{2I}{\mu_n C_{ox} {V_{od}}^2}$$

Substituting the values, we get $\frac{W}{L}_{4,5,6,7} \approx 10$, $\frac{W}{L}_{8,9,10,11} \approx 25$. For these transistors, $L = 0.6\mu m$

For M8-9, a self-biased circuit is employed to ensure the transistors operate in the saturation region while minimizing the use of additional bias circuits. For M4-7, a dedicated bias circuit is designed separately to guarantee that the corresponding transistors obtain the current required by the design.

Through simulations, we found that under the current parameters, the single-stage folded cascode amplifier exhibits an extremely high gain, but the bandwidth is insufficient—only about one-third of the design specification.

B. *Optimization and Adjustment*

The output bandwidth is mainly affected by the dominant pole, and reducing the $R_{out}C_L$ product is the most effective approach.

To increase the bandwidth, we attempted multiple strategies. First, we tried reducing the transistor size and adjusting device dimensions and model parameters to decrease the output resistance and node capacitance. However, the optimization effect was not ideal, resulting in only a minimal improvement in bandwidth.

Subsequently, we attempted to increase the bias current. Since $r_o \approx \frac{1}{\lambda I_D}$, increasing I_D can reduce r_o, thereby decreasing R_{out}. The tradeoff, however, is an increase in total current and a subsequent rise in power consumption. The effect was highly significant: when the bias current was increased to $I_3 \approx 120\mu A$,both the bandwidth and gain successfully met the design requirements.

As this is a single-stage amplifier, there are no stability issues. Therefore, there is no need to consider phase margin or design Miller compensation.

During the design process, the g_m/I_d method was not used for precise simulation and calculation of transistor parameters; instead, only formula derivation was relied upon. Consequently, due to the presence of secondary effects, there is a certain discrepancy between the actual operating parameters of the transistors and the designed values.

Subsequently, through repeated adjustments, attempts, and continuous optimization, a single-stage folded cascode amplifier that meets all design specifications was finally achieved.

VII. Single Stage Folded Cascode Amplifier Results

The overall schematic of the circuit is shown below, which clearly display all the design parameters.

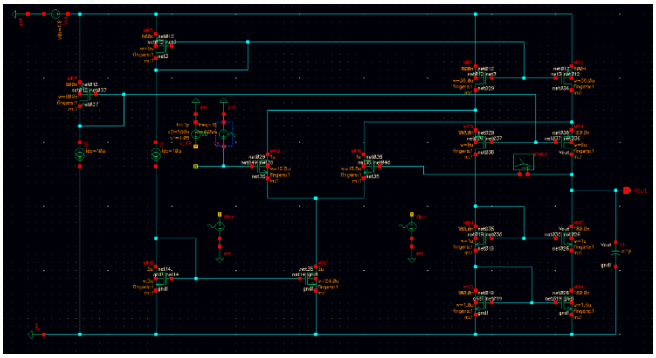


Figure 29. The Overall Circuit Schematic

Use the simulation tool ADEL, we can simulate our circuit, and the following figures show the result and performance of the two-stage 5-OTA based operational amplifier, under 1.8V 1.62V(1.8 × 90%) and 1.98V(1.8 × 110%) V_{DD}.

A. *Under 1.8V V_{DD}*

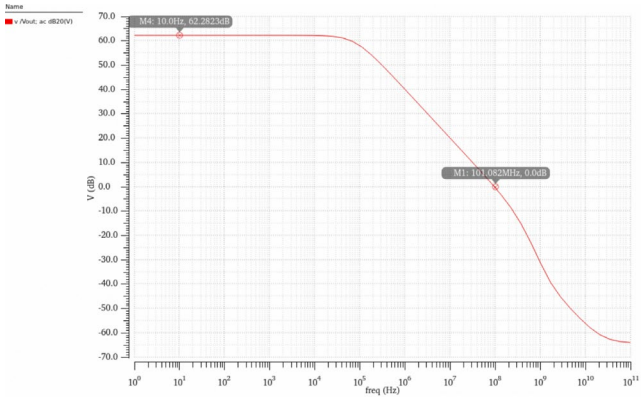


Figure 30. The Differential Mode Gain and Unity Gain Frequency

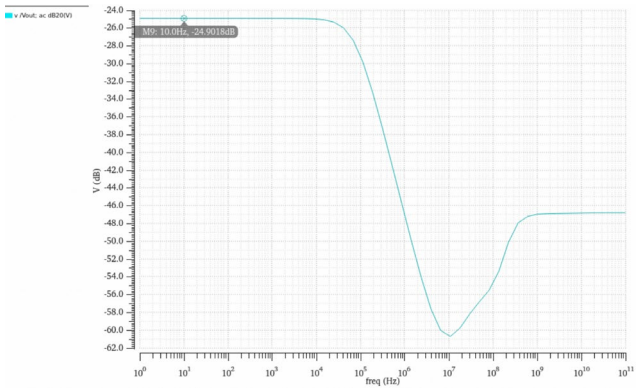


Figure 31. The Common Mode Gain

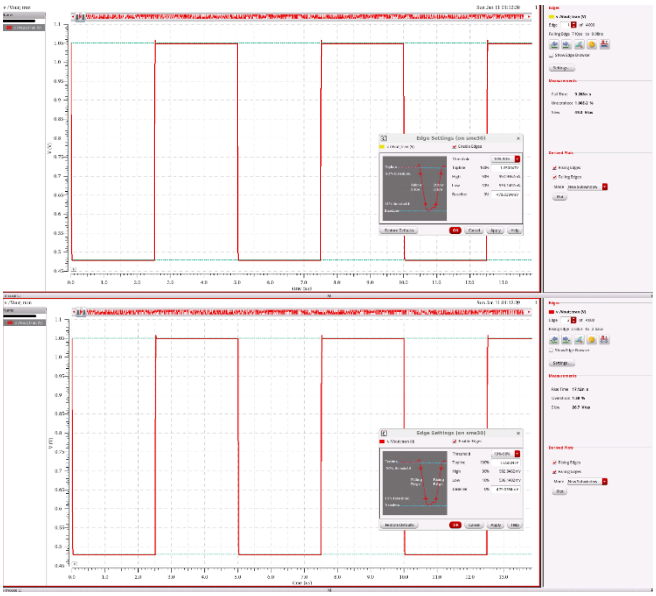


FIGURE 32. THE SLEW RATE

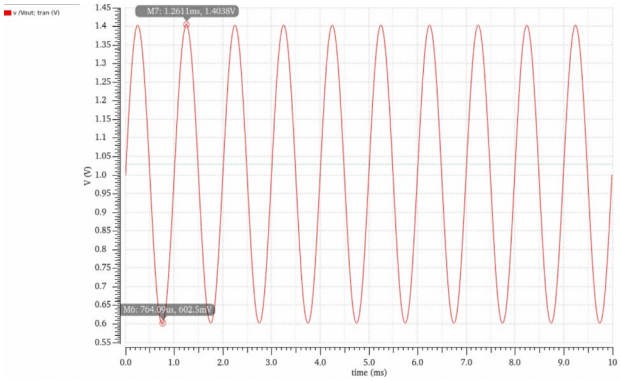


FIGURE 33. THE OUTPUT VOLTAGE SWING

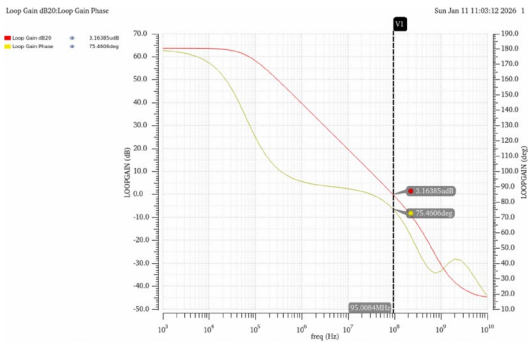


FIGURE 34. THE PHASE MARGIN

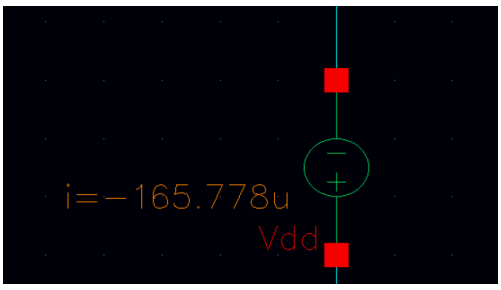


FIGURE 35. THE POWER CONSUMPTION WITH 20UA IREF

B. *Under 1.62V V_{DD}*

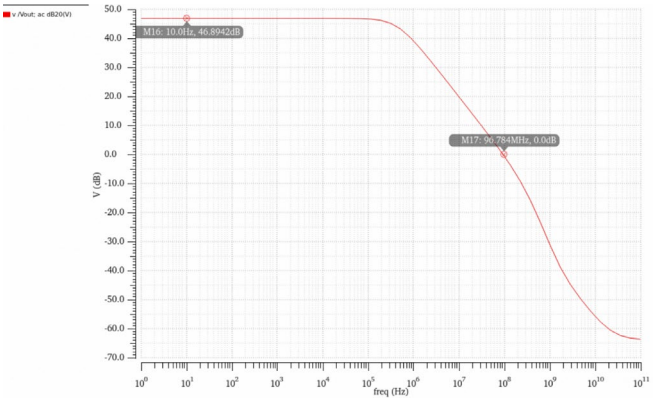


FIGURE 36. THE DIFFERENTIAL MODE GAIN AND UNITY GAIN FREQUENCY

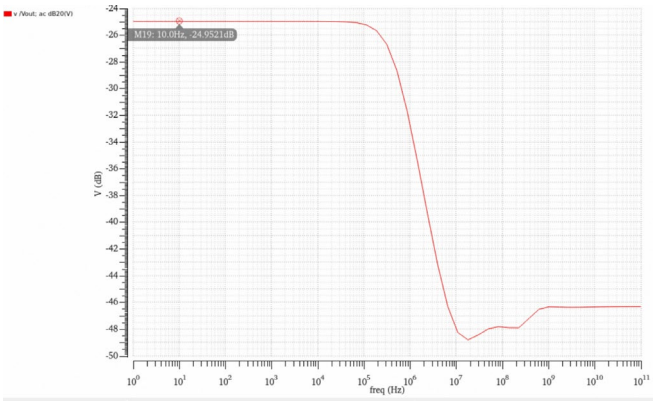


FIGURE 37. THE COMMON MODE GAIN



FIGURE 38. THE SLEW RATE

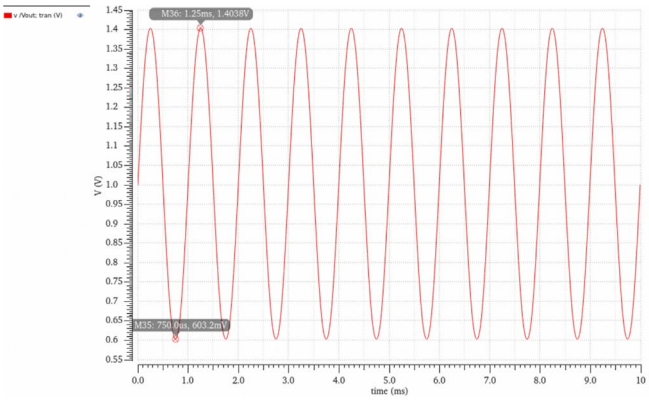


FIGURE 39. THE OUTPUT VOLTAGE SWING

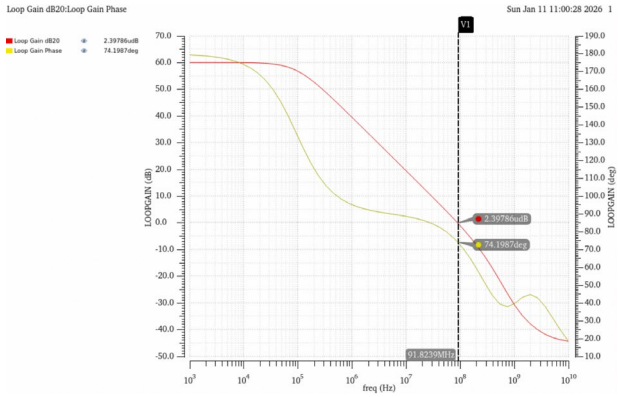


FIGURE 40. THE PHASE MARGIN



FIGURE 41. THE POWER CONSUMPTION WITH 20UA IREF

C. *Under 1.98V V_{DD}*

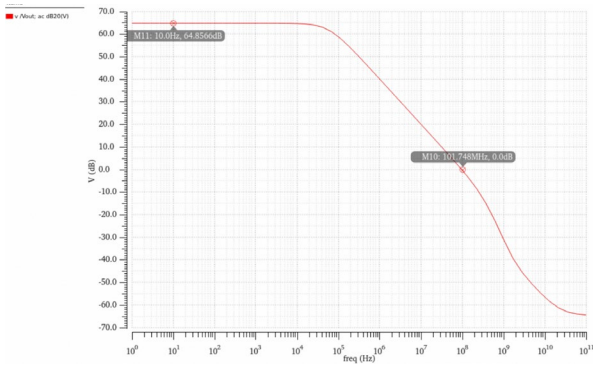


FIGURE 42. THE DIFFERENTIAL MODE GAIN AND UNITY GAIN FREQUENCY

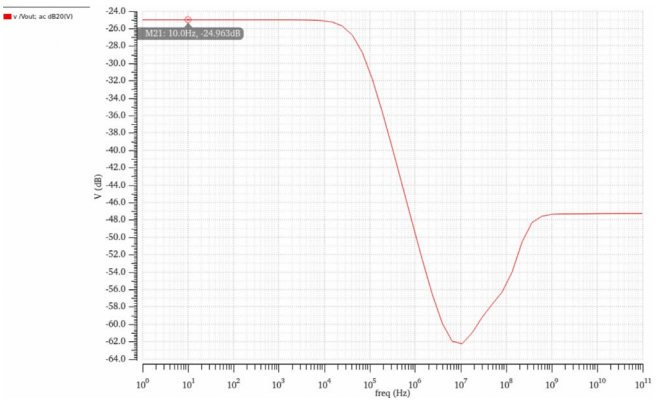


FIGURE 43. THE COMMON MODE GAIN

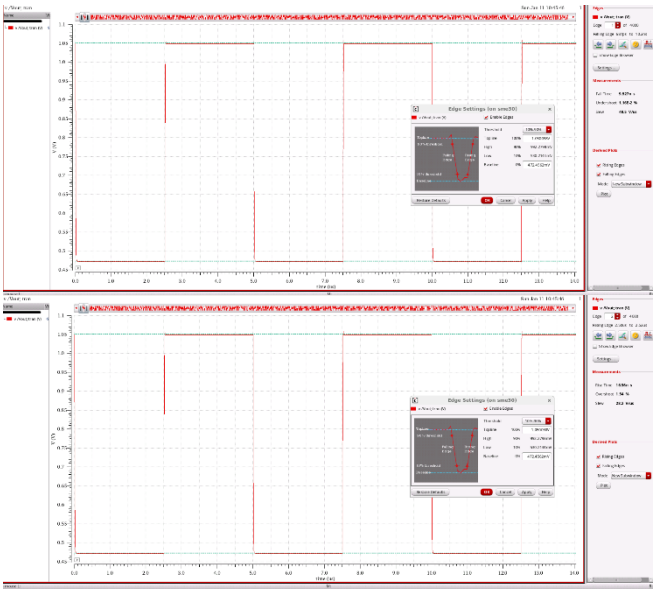


FIGURE 44. THE SLEW RATE

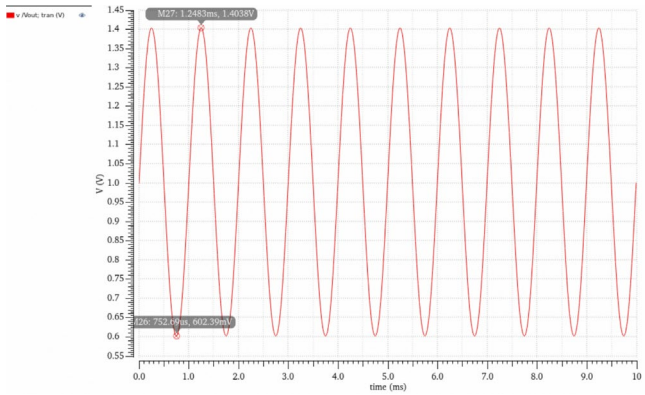


FIGURE 45. THE OUTPUT VOLTAGE SWING

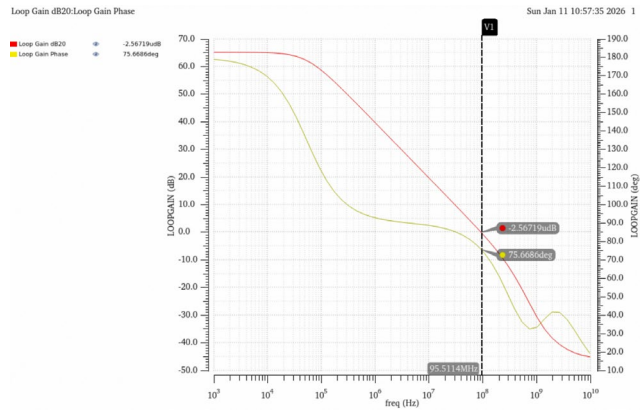


FIGURE 46. THE PHASE MARGIN



FIGURE 47. THE POWER CONSUMPTION WITH 20UA IREF

As can be seen from the above simulation results, all transistors operate normally in the saturation region under the three supply voltages. At 1.8V and 1.98V, the circuit simulation results fully meet the design specifications. However, at 1.62V, the decrease in overdrive voltage indirectly leads to a reduction in transconductance, resulting in a decline in both the circuit gain and bandwidth, which no longer satisfy the design requirements. However, it is foreseeable that as long as the power consumption is further increased, it will also be possible to meet the design specifications at a supply voltage of 1.62 V. Due to time constraints, no further attempts will be made in this regard.

VIII. CONCLUSION

This project successfully implemented two high-performance op-amp topologies that fully meet the design and bonus requirements.

The two-stage 5-OTA excels in stability and voltage fluctuation adaptability, with consistent gain, bandwidth, and phase margin across 1.62–1.98 V, though it consumes more power due to compensation components. The single-stage folded cascode offers simpler structure and lower power, with sufficient performance at 1.8 V and 1.98 V, but exhibits gain/bandwidth degradation at 1.62 V .

ACKNOWLEDGMENT

In This project, the contributions are equal, namely, 1:1. In detail, Yang Xianduo did the two stages 5-OTA amplifier, measured PMOS and NMOS basic parameters and proposed many useful tips, besides, Zhou Xiaojun did the single stage cascode amplifier and modified the reports.

The two stages 5-OTA structure amplifier is in the account panistc1, 307Lib library, final_project_display cell.

The single stage cascode structure amplifier is in the account panistc20, 307Lib library, folded_cascode cell.

REFERENCES

[1] B. Razavi, S. W. Director, Ed. Design of Analog CMOS IntegratedCircuits, 4th ed. United States of America: *McGraw-Hill Series inElectrical and Computer Engineering*, 2001.J. Clerk Maxwell, ATreatise on Electricity and Magnetism, 3rd ed.,P.J. H.

[2] https://www.bilibili.com/video/BV1Bv411e77i/?spm_id_from=333.1387.homepage.video_card.click&vd_source=f88cbcbcd75c5c61004ae2f749ac6e06